

H09 Agent swarms have an effective thermodynamics

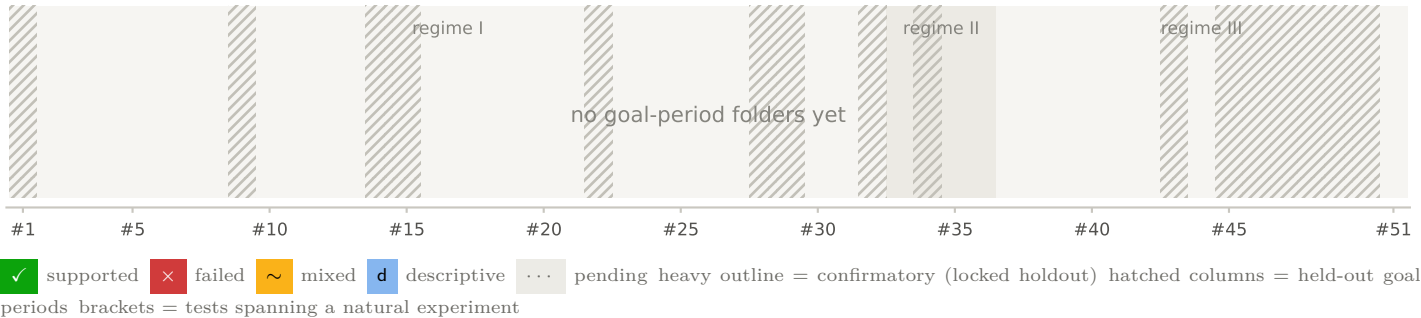
Complete 25% Faithfulness 1.5/5 Usefulness 2/5 level: hypothesis · pipeline checks 14% · updated 2026-10-04

exploratory rounds 1 (E1-E5) and 2 (E6-E8) done, light. Observable tables built (idle_runs, consolidation_inflow). Not promoted. Opened 2026-10-03 at Vivian's request.

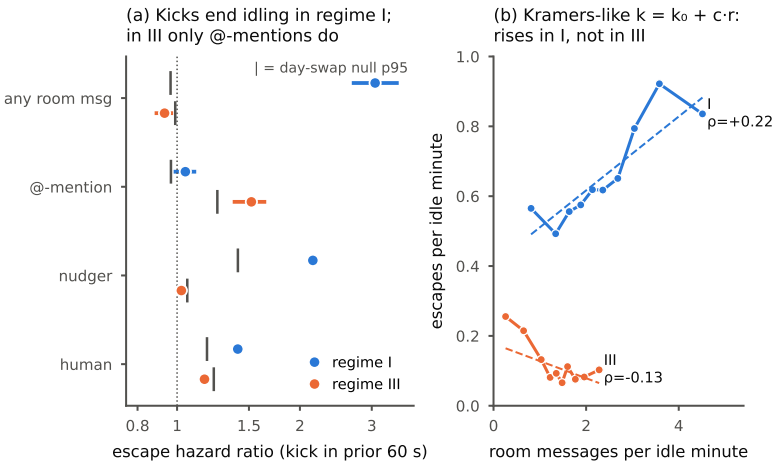
WHAT HAPPENED

An exploratory framing, not one test: can swarm statistics be read as thermodynamics (landscapes, currents, traps, equations of state)? Two light non-holdout rounds; no scorecard or holdout run. Round 1: the activity landscape's two wells are reproduced by the daily schedule (a field, not coupling). Per-agent action chains are irreversible (78% of cells beat the null), with a closed regime-III cycle consolidate → search → talk → pause, but event-level entropy production collapses after perma-computer-use. Agents are ergodic at day scale; restarts relax within a minute. Round 2 built reusable tables (idle_runs, consolidation_inflow). In regime I, room messages trigger escape from idling (hazard ratio 3.06; null p95 0.96). In regime III a pause is a self-set timer (0.22% end early), and only @-mentions matter (×1.52), at expiry. Memory size is an agent-specific set point, reset at each consolidation and barely moved by inflow. Collective entropy production: small (plug-in only).

GOAL PERIODS TESTED none yet



OBSERVABLES



PHYSICAL MODEL

No single model; each sub-claim borrows one (models 01, 02, 09, 10). $\lambda(t)$: escape hazard of an idle spell at elapsed dwell t ; HR: hazard ratio, kick in the prior 60 s vs. none. k_0 , c : base escape rate and kick sensitivity; r : kick rate. t^* : declared pause expiry; p_{gate} : chance of acting at expiry. π , P : stationary law and transitions of an agent's action-class chain. V : memory size; P_{in} : inflow; β , γ : response, relaxation.

$$\lambda(t) = P(\text{escape at } t \mid \text{idle until } t), \quad \text{HR} = \lambda_{\text{kick}} / \lambda_{\text{none}}$$
$$\text{regime I (Kramers-like): } k \approx k_0 + cr$$
$$\text{regime III (timer gate): } \lambda(t) \approx \delta(t - t^*) p_{\text{gate}}(\text{@-mention})$$
$$\sigma = \sum_{x < y} (\pi_x P_{xy} - \pi_y P_{yx}) \ln \frac{\pi_x P_{xy}}{\pi_y P_{yx}}$$
$$\Delta V = \beta P_{\text{in}} - \gamma V_{\text{prev}} \Rightarrow V^* = \beta P_{\text{in}} / \gamma$$

WHAT THE FIGURE SHOWS

(a) Escape hazard ratio of idle spells: minutes with a kick in the previous 60 s vs. minutes without, stratified by elapsed dwell and time of day. Bars: agent-day bootstrap CIs where computed; ticks: 95th percentile of a day-swap null. Regime I: any room message triples escape. Regime III: room messages don't help; only @-mentions do. (b) Escape rate against kick rate over agent-day deciles (dashed: linear fit $k_0 + cr$). It rises in I and falls in III.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The thermodynamic vocabulary did little work; the useful findings are plain mechanics. The one an operator can use: in the current scaffold (regime III), an idle agent is asleep on a timer it set itself. Room chatter does not wake it. Only an @-mention raises the chance it acts, and only when the timer expires. To get a response, address the agent by name and expect a delay of up to its chosen pause. Regime-I agents were interruptible. "Memory size is a set point" says consolidation is a reset, not growing storage. The rest restates the scaffold: landscapes made by the schedule, day-scale ergodicity, and a consolidation cycle that is the prompt's own loop. Nothing here yet beats a simpler statistic.

Bottom line. As a framing, swarm thermodynamics has not yet earned its keep; its best result is a practical fact about scheduling: in the current scaffold, idle agents are timer-gated, and only @-mentions get through.